

Find the area of a triangle with the given vertices.

7) $R = (3, -2, 7)$
 $S = (0, -7, 9)$
 $T = (6, -7, 9)$

8) $A = (6, 9, -8)$
 $B = (0, -9, -1)$
 $C = (-8, -9, 2)$

Find the area of a parallelogram with the given vectors as two adjacent sides.

9) $\vec{u} = \langle -7, 1, 0 \rangle$
 $\vec{v} = \langle 7, -5, -1 \rangle$

10) $\overrightarrow{AB}$ and $\overrightarrow{AC}$
Given: $A = (-8, 5, -8)$ $B = (4, 8, -8)$
 $C = (6, -3, 8)$

Find the volume of a parallelepiped with the given vectors as adjacent edges.

11) $\vec{u} = \langle -6, 4, -9 \rangle$
 $\vec{v} = \langle 2, -5, 8 \rangle$
 $\vec{w} = \langle 0, -4, 8 \rangle$

12) $\overrightarrow{TX}$, $\overrightarrow{TY}$, and $\overrightarrow{TZ}$
Given: $T = (0, -4, -8)$ $X = (8, 1, -9)$
 $Y = (-1, -5, -2)$ $Z = (0, -8, 3)$